

Q1 Choose the correct answer for the copper loss of a transformer

- [illegible]



Solve showing your work (10 Points)

A separately excited DC motor is rotated at 1000rpm, The variation of armature terminal voltage as a function of field current is measured under no-load conditions and table below:

I_f	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
V_T	0	30	60	85	102	115	124	130	134

The field winding supply 24V and the field resistance is adjustable. The armature winding resistance 0.2Ω and the armature terminal voltage is 130V.

- Calculate the field current if the motor is operated with no-load at 1000 rpm
- The motor drives a load at 1200 rpm. Calculate the armature voltage at 1200 rpm if the field resistance 60Ω
- Calculate the torque for the above condition

I_f

E_a =

T =



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Q3) Solve showing your work (10 Points)

A 240-V, 100-A DC shunt motor. This motor has an armature resistance of 0.14 ohm, field resistance of 200 ohm, and adjustable resistance with range from 0 to 300 ohm. With compensating windings and interpoles and field turn of 1500 turns. A magnetization curve for this motor at 3000 r/min can be summarized in the following table

I_f	0.1	0.3	0.48	0.5	0.6	0.75	1.1	1.2	1.3
E_a	60	140	195	200	220	245	278	282	285

- What is the no-load speed of this motor when $R_{adj} = 120$ ohm?
- What is its full-load speed?
- What is its speed regulation?

$n = \dots\dots\dots$ $E_a = \dots\dots\dots$

$SR = \dots\dots\dots$